



Assignment: Diagnostic Analysis using Python

Technical Report: NHS Appointment and Social Media (Twitter) Analysis

Context of the Business Scenario

The National Health services (NHS) in the United Kingdom is under huge pressure due to increasing population demand, workforce shortage, and the need to ensure efficient resource utilisation. The report examines NHS appointment data and social media data from Twitter (X) to evaluate patterns in demand, service delivery, and patient engagement.

The business problem in focus; is to determine whether the current staffing levels and infrastructure are adequate to meet the demand, and how effectively NHS resources can be used.

The analysis aims to answer two key questions:

1. Whether NHS capacity is sufficient
2. Whether resources are utilised efficiently

Moreover, the project explores whether external data sources also provide valuable insight into public sentiment. The analysis assumes consistent reporting across regions and time periods. The objective is to support data-driven decision-making to improve NHS efficiency, resilience, and long-term planning.

Analytical Approach

Data analysis is conducted using Python within a Jupyter Notebook that follows a structured analytical workflow consisting of data ingestion, cleaning, transformation, and analysis.

The primary datasets used are:

- `appointments_regional`
- `actual_duration`
- `national_categories`

Data ingestion is performed using Pandas functions (`read_csv`, `read_excel`), ensuring the datasets are loading correctly. Data structure and quality is confirmed by calling the functions such as `.info ()`, `describe ()`, and `head ()`. The datasets contain large-scale aggregated appointment data with minimal missing values, indicating strong data integrity.

Data wrangling includes removing duplicates, ensuring data types, and converting data variables into datetime format to support time-series analysis. The dataset is filtered to include records from August 2021 onwards, focusing on the relevant period to assess current NHS performance.

Aggregation is an important part of the analysis. The **groupby ()** function is used throughout to summarise appointment counts across variables such as healthcare professional type, appointment status, service setting, appointment mode, and waiting time. Daily utilisation, a derived metric is calculated by dividing monthly appointment totals by thirty. This helps with the NHS appointment benchmark of **1.2 million** appointments per day.

Visualisation is performed using Matplotlib and Seaborn. Line charts are used for time-series analysis, while bar charts are applied for categorical comparisons.

Visualisations are designed following best practices, including clear labels, appropriate scaling, and minimal clutter, to ensure clarity.

Furthermore, social media data has been analysed to assess its potential value. Tweets are cleaned by removing non-relevant entries, and hashtags are extracted and analysed using frequency distributions. This helps with the identification of key themes in healthcare-related public discussions.

A limitation of the method is that the datasets are aggregated, preventing data analysis at an individual patient level. Furthermore, social media data may not fully represent all population groups, potentially introducing bias.

Visualisation and Insights

Multiple visualisations have been generated to identify trends in NHS service demand and utilisation. Line plot display patterns over time and allow for comparison across multiple variables.

Monthly appointment trends show consistently high demand, with clear peaks during autumn and winter months. The seasonal variation reflects increased illness during colder periods.

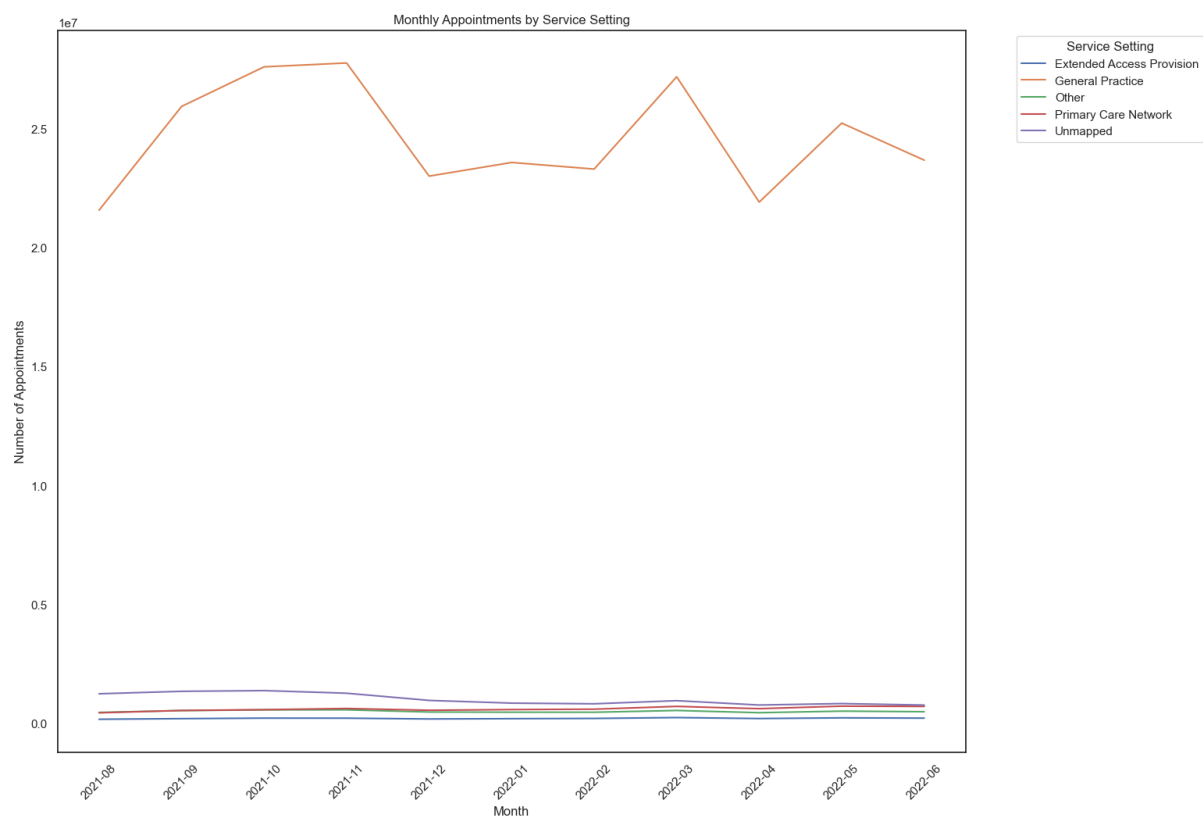


Fig 1. Monthly Demand Trend Line Plot

Capacity analysis indicates that daily appointment volumes approach approximately one million during peak periods, nearing the NHS benchmark of 1.2 million. The margin is narrow even when the capacity is not exceeded which depicts sustained operational pressure.

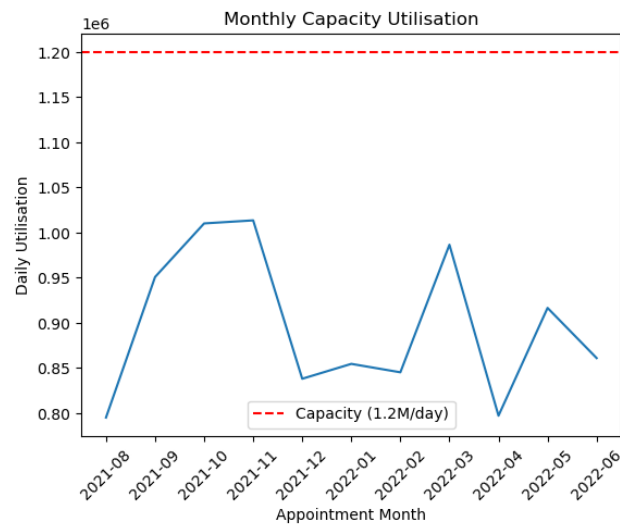


Fig 2. Monthly Capacity Utilisation

Analysis of healthcare professional types reveals a strong dependency on General Practitioners (GPs), although other staff contributes significantly. This shows a shift towards more distributed workforce.

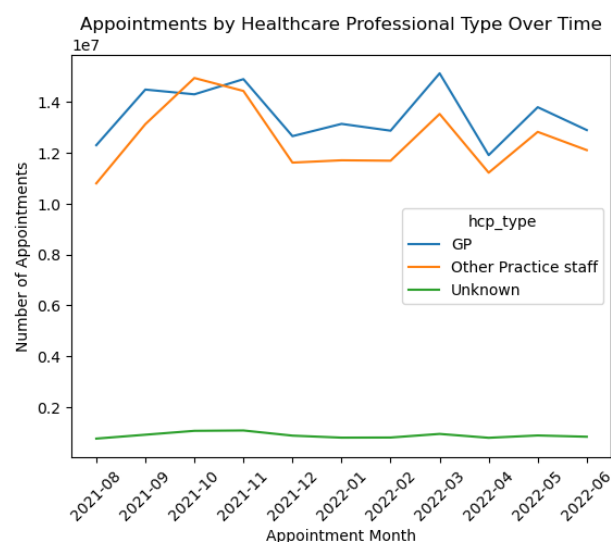


Fig 3. HCP Trend

Appointment attendance analysis shows that most appointments are attended. However, missed appointments (DNA) stay consistent, meaning there is inefficiency and list capacity.

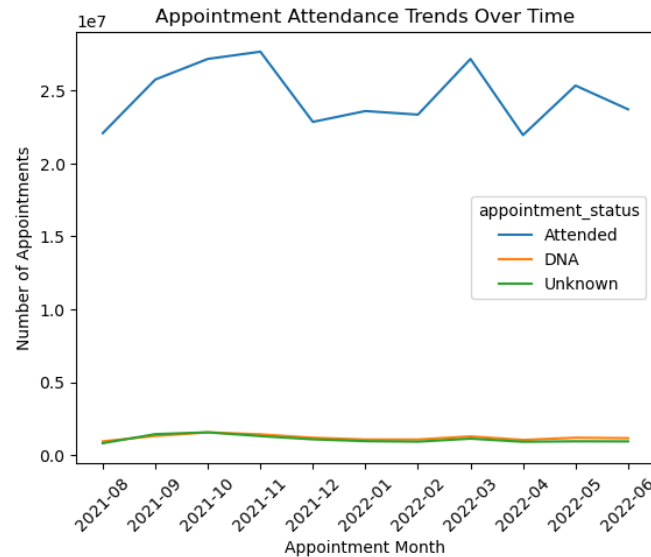


Fig 4. Appointment Attendance Trends

Appointment mode visualisation indicated that face-to-face and telephone consultations dominate service delivery while digital methods are underutilised.

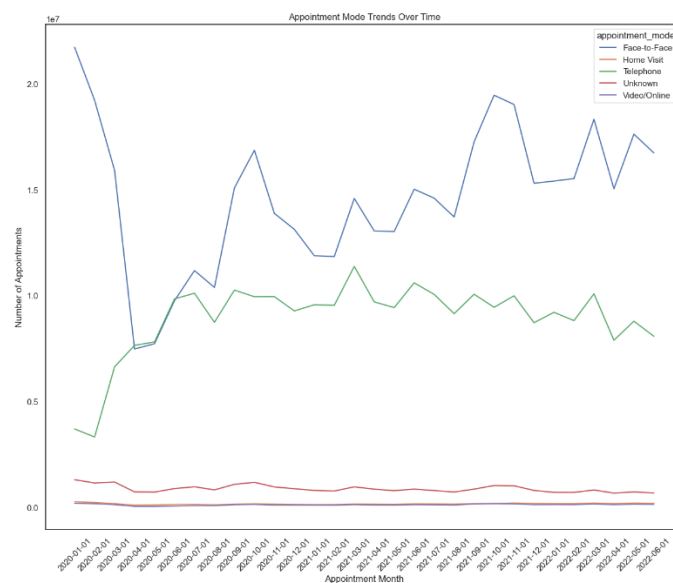


Fig 5. Appointment Mode Trends

Waiting time analysis shows that most appointments are booked on the same day or within one day, indicating robust performance in urgent care. However, wait times are long which means a delay in routine care.

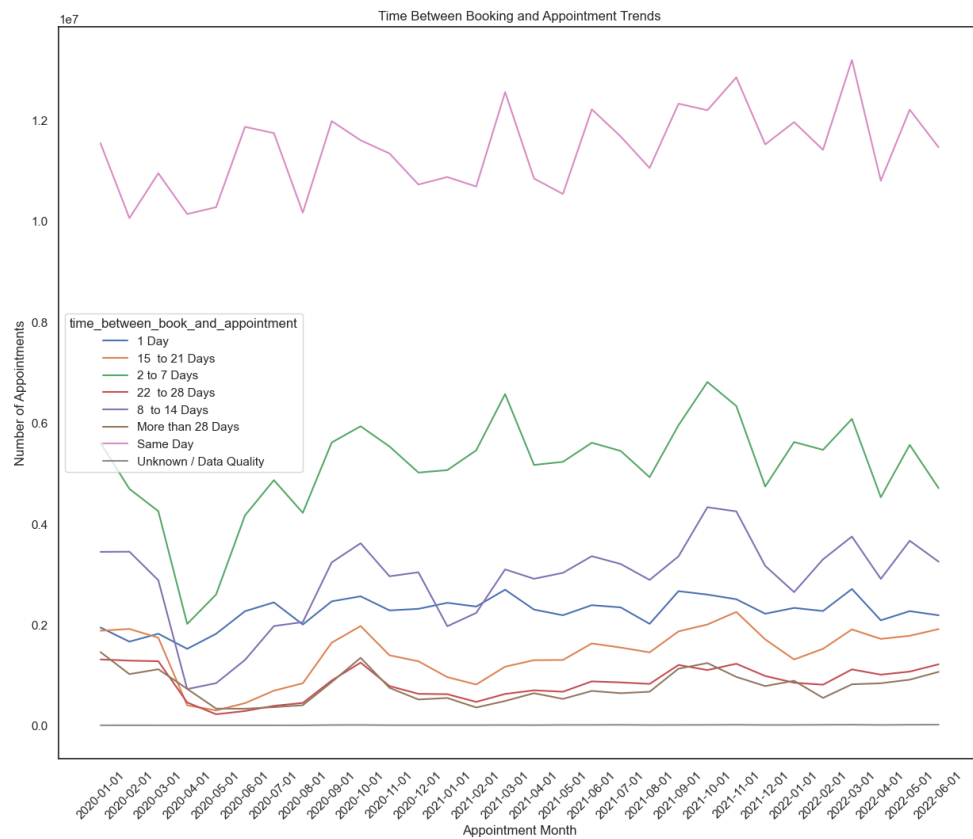


Fig 6. Time Between Booking and Appointment Trends

Service setting visualisation depicts a heavy reliance on GP services whereas alternate service settings are underutilised in comparison.

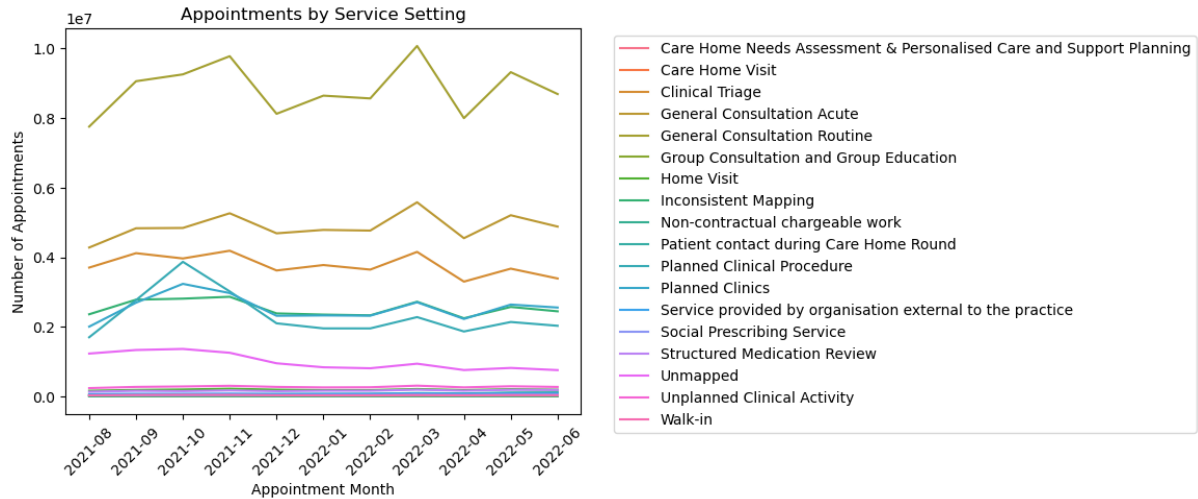


Fig 7. Appointments by Service Setting

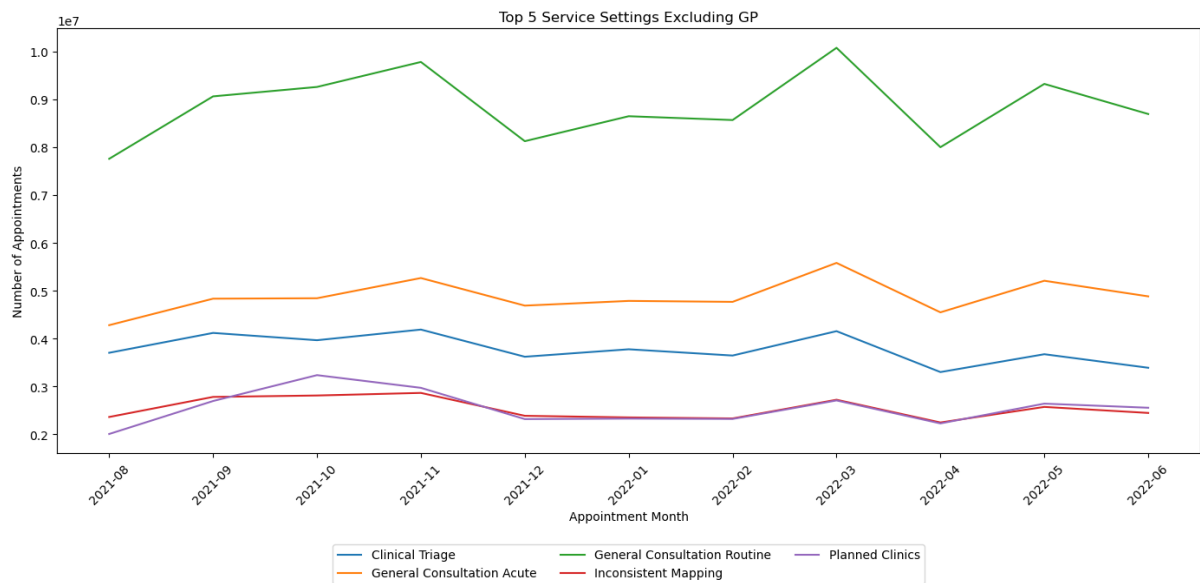


Fig 8. Appointments by Service Setting without GP

Social media analysis shows that #heathcare is the most dominant hashtag. It indicates a strong focus on general healthcare discussions.

Other hashtags such as #heath, #job, and #pharmaceutical show more specific themes which include workforce concerns and healthcare industry development; #digitalhealthcare hashtag suggests increasing public interest on digital transformation within healthcare.

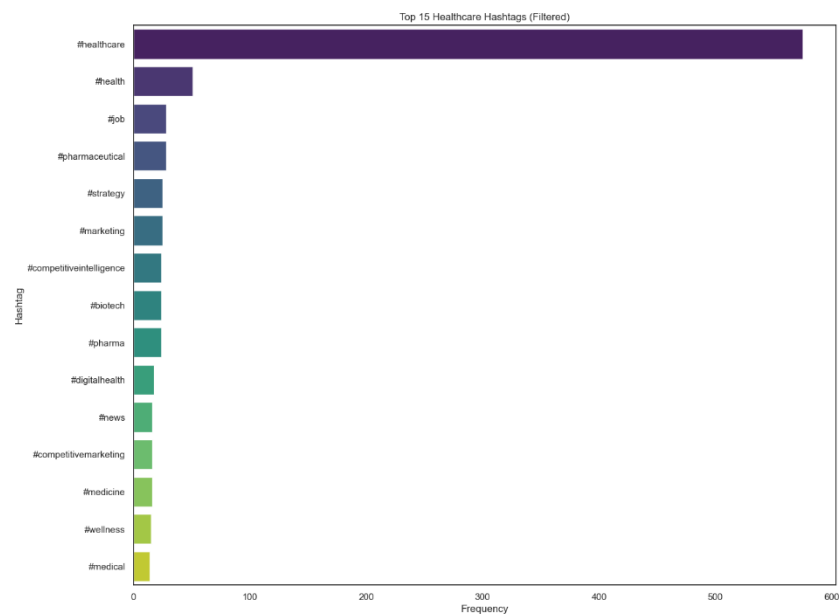


Fig 9. Hashtag bar chart

Patterns, Trends, and Recommendations

Data analysis reveals many key patterns and one of them is NHS services operating under high demand, with seasonal peaks during autumn and winter. Although capacity is not exceeded but the system operates with limited flexibility during peak time.

There is a strong reliance on GP services but there is also an increased shift in usage of other professional healthcare settings. Missed appointments show inefficiencies that reduce available capacity.

Waiting time analysis depicts impressive performance in urgent care delivery but identifies delays in routine services. Social media findings reinforce these insights by highlighting public concerns around workforce and access.

Recommendations:

This report identifies several opportunities for improving NHS efficiency, capacity, and service delivery:

- Increase workforce flexibility during peak periods by introducing flexible staffing models, such as temporary staff allocation or redistribution of resources, to better accommodate seasonal demand fluctuations and reduce pressure on services during high-demand periods.
- Expand the role of non-GP healthcare professionals and further develop multidisciplinary care models by increasing the use of nurses and allied health professionals to reduce reliance on General Practitioners and improve overall system efficiency.
- Reduce missed appointments (DNA) by implementing automated reminder systems via SMS or emails and improve cancellation processes.
- Expand digital healthcare services by increasing the adoption of video and online consultations to complement face-to-face and telephone services.
- Redistribute demand across service settings and encourage public to use alternative care pathways to balance and reduce dependence on GP services.

REFERENCES:

Artificial Intelligence (AI) tools were used to support code development, debugging, and visualisation refinement.

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